

Paper Modernization: Effects of Federal Policy to Insure Young Adults: Evidence from the 2010 Affordable Care Act’s Dependent-Coverage Mandate

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1 Introduction & Synopsis

1.1 Paper Introduction

This paper¹ studies the causal effects of the ACA’s dependent coverage mandate, which required insurers to extend parental health insurance eligibility to children up to age 26, on youngsters’ insurance coverage and labor market outcomes. The policy context is critical: young adults aged from 19 to 25 previously faced the highest uninsurance rates among all age groups (37% in 2008), partly because Medicaid eligibility phases out at 19 and employer-sponsored insurance (ESI) is typically bound to full-time employment. ACA created an age-specific expansion in eligibility, which the authors exploit in a DiD framework comparing newly eligible young adults with nearby untreated age groups before and after the reform.

The authors apply a difference-in-difference (DiD) framework using monthly SIPP 2008 panel data (covering Aug 2008-Nov 2011), comparing the treatment group (19-25 year-old individuals) to a combined control group of both younger and older ones (16-18 & 27-29) adults. The key outcomes variables include 5 insurance categories: any health insurance coverage (*anyhi*), employer sponsored dependent coverage through parents (*emphi_dep*), own-name ESI (*emphi*), individually purchased non-group coverage (*indiv*), and government provided coverage (*govhi*), as well as several labor market outcomes including full-time employment and weekly hours worked.

The empirical specification includes state and year FEs, individual-level demographic controls, state unemployment rates interacted with the treatment indicator, and separate dummy interactions for the enactment period (from Mar to Sep 2010) and the implementation period (since Oct 2010), the latter distinction being a methodological contribution compared to prior work that could not separate these phases with annual CPS data. The author furtherly used a triple difference (DDD) to focus on youngsters whose parents lacked ESI as a falsification group. They show that the mandate raised parental ESI by around 7.0% points and reduced uninsurance by 3.2% points during the implementation period, with the effect concentrated among those with lower marginal costs and higher marginal benefits of new coverage including non-students, non-married individuals, and whites exhibited significantly larger take-up than their counterparts. In addition, the authors also show evidence that, in the subsample with parental ESI, the ACA mandate cuts uninsurance rate by about 1/3, and it may increase labor market flexibility by reducing full-time work hours and weekly hours without affecting overall employment.

1.2 Identification Challenges

There are several identification and interpretation challenges that are especially relevant in this setting. First, since the main insurance outcomes are binary, the linear probability model (LPM) would conduct a limited approximation, as it treats treatment effects as linear and summarizes them through a constant marginal effect. However, the effects may vary with individuals’ baseline coverage and other factors. This concern is therefore solid by the paper’s subgroup and take-up analyses, which examine one-dimensional

¹See link.

heterogeneity at a time and therefore may miss more complex circumstances. Taken together, it raises concern that the original linear framework may simplify both the functional form and the dimension of heterogeneity.

Second, the parallel trends assumption. The original paper tests pre-trend by interacting the treatment indicator with a linear time trend (See Appendix Table A1) and finds no significant divergence. However, this approach only tests for a linear difference in pre-trends, and may ignore some period-specific deviations. Our quarterly event-study estimates suggest that, for the key outcome *emphi_dep*, some extreme pre-policy quarters show significant negative deviations (from 0) that are not captured. In addition, the original paper does not provide sensitivity analyses (joint test, event study, etc.) to show how robust its estimated ATE is to potential violations from parallel trends.

Moreover, it lies in the DDD specification in Table 5. Using samples whose parents were without ESI youngsters as a within-cohort falsification control is credible in design, and Appendix Table A4 confirms that parents' own ESI take-up was unaffected by the ACA mandate, validating the control group's stability. However, the specification still follows the LPM's functional form restrictions across three interaction dimensions, making it difficult to distinguish a null in the falsification group from specification error. Finally, the take-up analysis in Table 6 (original paper) uses bivariate t-tests that can not rank joint predictive power of multiple friction sources.

1.3 Modernization Models Used & Report Structure

The modernization report does not try to overturn the paper's central conclusion that the ACA dependent-coverage expansion increased parental ESI take-up. Instead, it makes three contributions. First, it reassesses the rigor of the identifying assumptions using dynamic event study evidence through HonestDiD analysis, and an age placebo test. Second, it re-examines the paper's key average treatment effects (ATE) using semiparametric robustness checks, including DRDID and DRDDD. Third, it extends the original subgroup and take-up analyses by studying treatment effect heterogeneity with causal forest and non-takeup prediction with a random forest, which shares the same logic. For classification, latter models are interpretive and predictive extensions rather than replacements for the original design.

They collectively construct the remainder of this paper. Section 2 addresses limitations through Doubly Robust DiD, Causal Forest, Event Study with HonestDiD sensitivity analysis, and Doubly Robust DDD. Section 3 conducts causal forest heterogeneity analyses, documenting an education gradient in CATEs that the original paper does not examine, as well as other dimensions. Section 4 conducts a permutation-based age placebo test that complements the original temporal placebo. Section 5 extends the take-up analysis using a random forest classifier. Section 6 concludes.

2 Modern Identification Strategies

In this section, we apply different modernized models to solve different estimation bias or challenges. For each approach or model, we first state the identification challenge to anchor, followed by a brief introduction of the principles and their contributions, as well as the relationships with this paper. After that, we introduce our approach to use that model, then compare and interpret the results generated by the model and the original ones shown in the modernized paper. At the end of this section, I will briefly explain the reason why I do not choose, or choose to exclude using some potential modernization models.

2.1 Doubly Robust DiD

The original LPM DiD estimates a constant marginal effect across all individuals, even though responses may vary with baseline coverage and other covariates. However, the LPM assumes a linear function form

and homogeneous marginal effects, which may impose limitations for binary insurance outcomes. In order to test whether the baseline are sensitive to these assumptions, we use Doubly-Robust DiD (DRDID) here as a semiparametric robustness check, which integrates outcome regression with propensity score weighting. The key advantage of DRDID is double robustness, as the estimator remains consistent if either the outcome model or the propensity score model is correctly specified. Therefore, it provides a useful robustness check on whether the baseline levels are sensitive to linear functional form assumptions or covariate unbalance among treatment and control groups.

Sant’Anna and Zhao (2020)’s Doubly-Robust DiD, unavailable to the original paper, addresses this by combining outcome regression with propensity score weighting, remaining consistent if either component is correctly specified. It estimates the average treatment of the treated (ATET), see equation 1 below:

$$ATET = \mathbb{E}[Y_i(1, 1) - Y_i(0, 1)|D_i = 1] \quad (1)$$

under the (conditional) parallel trend test assumption. Therefore, the DRDID estimator combines the outcome regression $\mu_{d,t}(X_i)$ with a propensity score $\pi(X_i) = P(D_i = 1|X_i)$. To fit that structure, to implement it, within each period window, we collapsed observations to individual level means before reconstructing them into the two-period panel structure² required by drdid. during the sample month for the enactment and implementation periods. More importantly, this changes both the data structure and the estimands, so result comparisons should be interpreted cautiously.

Technically, this semiparametric approach relaxes the functional form restriction in LPM, while keeping the traditional DiD identification idea. However, there are 2 caveats needed to be paid attention to: Firstly, DRDID focuses on ATET rather than ATE, which is different from the original estimate (they have different magnitude, etc.), when treatment effects are heterogeneous across the age distribution. Second, collapsing the data will eliminate within-person variation during the sample months.

We compare the original results with DRDID. If the DRDID and original LPM estimates are similar, at least on the same direction, the main finding is unlikely to be deviating by the LPM’s functional form restriction. If they differ substantially, the discrepancy should be interpreted with cautious, because it may reflect not only form sensitivity, but also differences in estimands or reweighting.

Table A.1 reports the DRDID ATET estimates when comparing the original LPM DiD estimates. For the outcome *emphi_dep*, the DRDID ATET during the enactment period is 6.31% ($p < 0.05$), consistent in direction to the original 2.39%, though somewhat larger in magnitude. It is consistent with the ATET which focuses on the treated subpopulation where take-up responses are concentrated. During the implementation period, the two estimates converge further: DRDID yields 7.33% points against the original 7.02%, both are significant. It suggests that, for *emphi_dep*, we can not show evidence that the original finding is primarily driven by the limitations from LPM’s functional form.

For *indiv* and *emphi*, the DRDID estimates are relatively small and mostly insignificant, but some of them are inconsistent with original paper’s in terms of directions or signs. For *govhi*, the DRDID estimate is positive and significant during both periods, diverging from the near-zero or negative estimate in the original paper.

The *anyhi* results are worth a separate discussion. The DRDID ATET for *anyhi* is large and positive in both periods (enactment: 11.76%; implementation: 11.43%), diverging from the original’s insignificant enactment effect and relatively modest (3.18%) implementation effect in larger scale. We suggest that, the larger difference of *anyhi* should not be considered as an evidence that the original estimate is incorrect. Instead, it suggests that the DRDID estimate for this outcome should be interpreted cautiously in this

²It means that, for each individual, we only keep one pre-policy and post-policy each, dropping individuals who have only one-period observation.

setting, because the large divergence may reflect reweighting, estimand differences, and substantial pre-policy differences across age groups, making it less comparable.

2.2 Causal Forest as a Semiparametric Extension

The main purpose of causal forest in this section is not to replace the paper’s DiD design with a new baseline estimator, but to assess whether the original ATE framework can mask meaningful treatment effect heterogeneity. The paper’s LPM estimates a single ATE, assuming the observations in all treatment group (19-25-year-old) as homogeneously affected by the ACA. However, the paper itself assumes that the take-up should be concentrated among those who with lower costs and higher benefits of getting new coverage. Compared to traditional OLS estimator, it can identify individual-level heterogeneity rather than a simple ATE. Furthermore, causal forest requires a stronger identifying assumption than the original LPM DiD.

Given that there’s 5 outcome variable in Table 2, we only pick 2 of the most important ones in this research topic for simplicity, which are the *emphi_dep*, depicting the employer dependent coverage. In addition, the core outcome *anyhi*, indicating whether the individual has health insurance anywhere.

Our approach is that, employing causal forest³ (Athey and Imbens (2019); Wager and Athey (2018)), a post-2013 method that provides an honest CATE estimates without pre-specifying subgroups, directly addressing the multiple testing and within-group conflation problems of the original Table 4 subgroup regressions to estimate CATEs (See equation 2) under conditional unconfoundedness (See equation 3). Methodologically, causal forest uses recursive partitioning to treatment effect estimation by searching for covariate splits that reveal systematic heterogeneity in causal responses, while using “honest” sample splitting approach to reduce overfitting (Sylvia et al. (2021)).

Because causal forest is designed for cross-sectional data, I construct two transformed cross-sectional settings⁴, CF1 and CF2. CF1 uses the last post-policy observation for employer dependent coverage (*emphi_dep*), because the policy is more fully in effect later in the implementation window (Table 3 in the original paper also agrees with that). This specification is to study heterogeneity rather than to replace the paper’s original DiD design. CF2, instead, uses a pre-post difference outcome ($\Delta Y_i = Y_{i,post} - Y_{i,pre}$) for *anyhi*, where baseline age profiles for *anyhi* make a pure cross-sectional interpretation less convincing, transforming the identification assumption from cross-sectional unconfoundedness to an assumption similar to parallel trend that has validated in the original paper’s Appendix Table A.

$$\tau(x) = E[Y_i(1) - Y_i(0)|X_i = x] \quad (2)$$

$$\{Y_i(0), Y_i(1)\} \perp W_i | X_i \quad (3)$$

$$Y_i(0) \not\perp W_i | X_i \quad (4)$$

Table A.2 compares the causal forest ATE estimates and the original LPM estimates. For *emphi_dep* under CF1, the implementation-period ATE is 1.9% ($p < 0.01$), directionally consistent with the original 7.0% LPM estimate but smaller in magnitude. This result is expected given that the cross-sectional collapse captures a narrower slice of the implementation window and targets a population-average effect under conditional unconfoundedness. The insignificant enactment-period estimate (0.005, $p = 0.46$) is consistent with

³The causal forest in this section is used primarily as an extension to assess treatment effect heterogeneity beyond the paper’s ATE framework, and secondarily as a supplementary comparison for ATE under a more flexible estimation.

⁴These approaches should be viewed as semiparametric extensions under outcome-specific identifying assumptions, rather than as direct replacements for the paper’s original DiD estimator.

the original’s finding that anticipatory *emphi_dep* take-up was modest relative to the implementation effect.

2.3 Event Study & Honest DiD

The original paper tried to verify parallel trend using a parametric pre-trend test (At Appendix Table A1), Which follows the standard DiD framework using pre-enactment as pre-policy periods. It concluded that there is no statistically trend difference exists between treatment and control groups prior to March 2010. However, this test has two limitations. First, it imposes simple linearity on the pre-trend, potentially masking non-linear deviations at specific time points. Second, because the original paper reports only two post-policy periods, it does not reveal the full dynamic path of the treatment effect.

To avoid the sample imbalance in periods far away from March 2010 and take fully advantage of the entire dataset, we complement the analyses with a quarterly event study, trying to estimate:

$$Y_{igst} = \alpha + \sum_{k \neq -1} \beta_k * D_i * [rel_qt = k] + X_{igst}\gamma + \tau_t + \zeta_s + \epsilon_{igst} \quad (5)$$

where rel_qt denotes quarters relative to 2010Q1 (base period = 2009Q4), $D_i = \mathbf{1}[\text{age} \in [19, 25]]$ is the treatment indicator, and τ_t, ζ_s indicate quarter and state fixed effects. The coefficients $\{\beta_k\}$ trace the dynamic treatment effect across quarters, with $\beta_k = 0$ for all $k < 0$ constituting the parallel trends assumption.

Figure A.1 presents event study estimates for *emphi_dep* and *anyhi*. *anyhi* does not show evidence for violating the parallel trend assumption. But for *emphi_dep*, however, the more distant pre-periods (2008Q3–2009Q1) show several negative deviations, raising concerns about whether the standard event-study estimates may overstate the post-policy effect. If these early deviations reflect a genuine divergence between treated and control groups, then the estimated increase after the ACA could be upward biased. If instead they are driven by the rotational structure of the SIPP, which interviews only one quarter of the sample each month and can generate uneven quarter-level composition, and they are not considered a violation of parallel trends.

However, a standard event study inference assumes that the parallel trends assumption holds in each post-period, even when pre-period violations are detected. To deal with that, Rambachan and Roth (2023) propose a partial identification approach: HonestDiD, which "honestly" relaxes this assumption by bounding the magnitude of post-period violations and reporting confidence intervals that remain valid under the given bound.

The identification framework is the following: let $\delta_t = E[Y_{it}(0) | D_i = 1] - E[Y_{it}(0) | D_i = 0]$ denote the violation of parallel trends in period t . While standard DiD assumes $\delta_t = 0$ for all t , HonestDiD relaxes this assumption by imposing the Relative Magnitudes (RM) restriction:

$$\Delta^{RM}(\bar{M}) = \left\{ \delta : \forall t \geq 0, |\delta_{t+1} - \delta_t| \leq \bar{M} \cdot \max_{s < 0} |\delta_{s+1} - \delta_s| \right\}. \quad (6)$$

which bounds the maximum post-policy period violation to $\bar{M}$ times the largest pre-policy period deviation. As $\bar{M}$ increases, CI expands to allow larger potential violations. With the expansion of CI, it will start to cover zero, thus the breakdown value $\bar{M}^*$ at the smallest $\bar{M}$. A high $\bar{M}^*$ constitutes a passing result, indicating robustness; the relatively low $\bar{M}^*$ under all pre-periods therefore represents a marginal robustness, suggesting the original estimation is vulnerable to slight post-period violations.

Based on that, we conduct two diagnostic steps to assess the severity of the parallel trends violation for *emphi_dep*. First, I test how sensitive the estimated policy effect is to the scale of post-period violations.

Second, we argue whether the violation is driven by the distant pre-periods (pre7-pre5 in this case, corresponding to 2008Q4-2009Q1) or reflects a trend difference that persists in the near pre-policy periods. If the former, restricting the pre-period window to the quarters immediately preceding the mandate, where pre-trends are flat, should yield more robust inference. These two questions rely directly on the HonestDiD sensitivity framework: the breakdown value $\bar{M}^*$ answers the first question, and its comparison across full versus dropping the extreme periods answers the second one.

The results are the following. For *emphi_dep*, Figure A.2a report the HonestDiD results. When using all pre-periods, the standard CI at $\bar{M} = 0$ is $[0.013, 0.046]$, excluding zero. The breakdown value is $\bar{M}^* \approx 0.15$, indicating that the conclusion is sensitive to post-period violations exceeding 15% of the maximum pre-period deviation. This threshold reflects the influence of distant pre-period deviations (pre6: -0.046, pre5: -0.038), which may be driven by composition-related differences in the SIPP rotation sample rather than pure pre-trends.

Comparatively, when restricting the pre-policy period into pre4-pre2, the breakdown increase significantly to $\bar{M}^* \approx 0.5$, as Figure A.2b shows. The parallel trend, correspondingly, can be satisfied within this restriction (See Figure A.2c). It indicates that the policy effect remains positive unless post-period violation reaches 50% of the violations among pre-policy periods. Overall, these results suggest that the policy effect on parental ESI take-up is robust to moderate departures from parallel trends.

2.4 Doubly-Robust DDD

In Table 5, the original DDD specification introduced a 3rd dimension: the parental ESI status. Essentially, this approach still follows the LPM’s functional form restrictions and solely applies them into 3D interaction without solving the misspecification compounds. The LPM’s constant effect assumption is particularly restrictive in the DDD context, where the relevant comparison group (*emphi_parent* = 0) should produce a non-significant effect, and any deviation from zero in the LPM estimate may reflect specification error rather than a pure policy effect.

Using DRDDD, we replace the previous DRDiD into DRDDD by separating *emphi_parent* with 1 and 0 subgroups. This design follows the same logic as traditional DDD by using youngsters without parental ESI access as a within cohort falsification group. The *emphi_parent* = 0 group provides a particularly clean falsification test since zero without parental ESI, any nonzero ATET in this group would indicate a confounding trend rather than a pure policy effect.

Table A.3 reports the DRDDD estimates⁵. For the treated subsample, the *emphi_dep* ATET is 14.8% points ($p < 0.01$), larger than the original’s 9.6% points. The result is consistent with when propensity score reweighting placing greater weight on compliers which has relatively lower proportion. However, the falsification subsample is less informative for this outcome, because *emphi_dep* is unavailable when there is no parental ESI access. Therefore, the main value of the DRDDD result for *emphi_dep* is confirmatory: it shows that the positive treatment-side effect remains under a less parametric specification.

For other outcomes, *anyhi* ATET of 8.2% points ($p < 0.01$) and the significant negative *emphi* effect (-4.5%, $p < 0.05$) are both directionally consistent with the original, suggesting the main DDD conclusions are robust to if we change the function form to semiparametric one.

Overall, the DRDDD results support the original DDD conclusions in the following ways. First, the treated sample estimates for *anyhi* and *emphi_dep* remain positive and economically meaningful. Second, the negative effect on *emphi* is consistent with insurance type substitution from own-name employer-sponsored insurance toward parental coverage. Taken together, these results suggest that most of the DDD conclusions are robust to relaxing the linear probability specification, especially for core outcomes *anyhi* and *emphi_dep*.

⁵Due to the length limit, we only analyze implementation effect in this section.

The contribution of the DRDDD is therefore primarily confirmatory, as it strengthens the evidence that the original findings are not mainly driven by the functional form assumptions of the LPM DiD.

2.5 Excluded Models and Justifications

There are several estimators have been considered by ultimately excluded. First, we abandoned Synthetic DiD (Arkhangelsky et al. (2021)), and *did_multiplegt* (De Chaisemartin and D’Haultfoeuille (2024)), both require unit-level treatment variation over time. In this setting, however, the ACA mandate was implemented nationally, and identification comes from age-based treatment variation within periods rather than staggered policy adoption across states. For that reason, these estimators are less well suited to the paper’s design. Second, the CSDiD (Callaway and Sant’Anna (2021)) faces the same limitation, since it does not have variation in treatment timing across groups, whereas the policy was implemented at the same date. Comparatively, a standard event study and sensitivity analyses fit the identification problem in this paper. Last but not least, Double Machine Learning in DiD (DMLDiD by Chang (2020)) was ruled out because the deterministic treatment assignment leads the propensity score component to degenerate.

3 Heterogeneity Analyses

Given the data limitations that reduce the rigor of heterogeneity analysis for *anyhi*, this section only focuses on *emphi_dep*, the outcome that most directly captures take-up of parental employer-sponsored insurance among eligible young adults. Heterogeneity in *emphi_dep* CATEs therefore provides a useful complement to the ATE analysis in Section 2 by showing how take-up responses vary across observable factors.

The original paper has already conducted comprehensive subgroup DiD regressions across all socioeconomic characteristics, so we choose not to replicate those results here. But all analyses are one-dimensional. Therefore, we focus more on the dimensions that the causal forest approach can contribute other than the original paper’s analyses, including the education gradient (and other dimensions), which the original paper does not examine, and the temporal changes of heterogeneity for enactment and implementation periods, which the original paper’s period subgroup estimates cannot capture.

According to Sylvia et al. (2021), DiD subgroup regression for heterogeneity approach still faces two limitations. First, testing subgroups one by one raises concerns about multiple testing and over-rejection, while low-dimensional subgroup splits may ignore important sources of heterogeneity. We therefore use causal forest, not to replace the original DiD design, but to provide a more data-driven, and still interpretable way to identify which baseline characteristics are most strongly associated with treatment-effect heterogeneity.

In terms of the cross-sectional data construction, we follow the steps in Section 2.2 above. To separate enactment to implementation effect, we limit the post-period as: later than Oct 2010 (while excluding the enactment period) and only keep Mar-Sep 2010. After estimating the Causal Forest ATE, we use the Best Linear Projection (BLP) function onto the covariate collection to estimate if a certain variable can contribute to data heterogeneity.

3.1 Best Linear Projection (BLP)

To statistically identify which characteristic(s) can contribute more on heterogeneity, we estimate it by using Best Linear Projection (BLP) approach of individual CATEs onto the covariates, which is available for *best_linear_projection* function in *grf* function in R. The BLP summarizes which characteristics can linearly predict the estimated treatment effect heterogeneity estimated by the causal forest. Formally, the BLP estimate:

$$\tau(X_i) = \alpha + \beta' X_i + \epsilon_i \quad (7)$$

where β captures the linear relationship between each covariate and the individual CATE, with standard errors that are cluster- and heteroskedasticity-robust. A significant coefficient indicates that this factor can predict treatment effect heterogeneity. It's a stronger way to detect heterogeneity compared to the original paper's subgroup DiD estimation. Table A.4 reports selected BLP results⁶ for the CF1 specification across settings including both enactment period and implementation period. The main result is that, education-related variables are the most persistent predictors of heterogeneity, especially during enactment, while marital status and student status become more important during implementation. By contrast, female and white are not statistically predictive in either period.

We suggest that, from the BLP results, the main findings are significant from education gradient (different level of degrees). The heterogeneity is strongest among HS Dropouts and HS Graduates (significant) and weakens as education rises, with College Grads significant only at enactment and insignificant by implementation. It agrees with individuals with lower degree that are more likely to be lacking access to employer sponsored coverage, making parental plan availability the main determinant of their insurance status. In comparison, college graduates are more homogeneously covered through other channels, leaving little heterogeneity. The temporal change suggests that, from policy enactment to implementation, education-related heterogeneity diminishes, while household-related heterogeneity (incl. married, student status) appears, indicating that alternative coverage sources start to become the main constraint.

The calibration test⁷ in Table A.5 shows that: the calibration test confirms treatment effect heterogeneity in all three specifications: the differential forest prediction is significant at the 1% level for both implementation-period forests (CF1: $\hat{\beta}=1.126$, $p < 0.001$; CF2: $\hat{\beta} = 1.182$, $p < 0.001$) and at the same level for the enactment period (CF1: $\hat{\beta}=0.541$, $p = 0.008$).

3.2 Education Gradient

One of our new findings from Causal Forest model is education gradient (The original paper did not report this result) both during the enactment and implementation period, based on the BLP estimation. Figure A.3a and A.3b visualize the BLP results by boxplots. During the enactment period, the CATE distribution is compressed near zero in aggregate, yet education-group boxplots reveal a clear gradient: Some College and College Grad show more positive median CATEs, while HS Dropout remains near zero. It suggests that more educated young adults can respond earlier to the new coverage option because they face lower information, administrative, and enrollment costs. During the implementation period, the pattern remains non-monotonic, as Less than HS retains a relatively high median CATE, whereas HS Dropout keeps below the ATE line. This suggests that education-related heterogeneity reflects not only differences in take-up frictions, but also possible differences in the gains from coverage.

3.3 Other Heterogeneity

We estimate that there are 3 additional dimensions that show different scale of heterogeneity during enactment and implementation. Young adults with no prior dependent coverage (*emphi_dep_pre* = 0) exhibit higher implementation-period CATEs than those already on parental coverage. Because if someone already had dependent coverage before ACA mandate, it has less space to rise, that is, insurance inertia limits the

⁶For other covariates, they are insignificant neither in enactment or implementation period. We choose not to show them due to length limit.

⁷Following Sylvia et al. (2021)'s code, a necessary prerequisite of a solid BLP test is that the heterogeneity is indeed detectable. To test that, we follow the calibration test proposed by Chernozhukov et al. (2018), which is also available at *grf* package. A significant coefficient on the differential forest prediction will reject the null hypothesis of treatment effect homogeneity.

marginal response among pre-existing enrollees (See Figure A.4). Married individuals exhibit higher CATEs than single individuals ($mar=0$) during implementation (See Figure A.5). Non-students ($ft_student$) exhibit higher CATEs than full-time students during implementation (See Figure A.6), reflecting students' greater pre-ACA access through prior IRS rules.

Compared to Table 4 in the original paper, we find the following results: First, we both find that the effect is higher in non-students than in full-time students, showing a significant difference, especially in implementation period. The CATE boxplot (Figure A.6) also agrees with that. Second, however, marital status is less straightforward. Although the BLP coefficient on mar is statistically significant $[-0.059 (0.018)]$, the boxplots do not display the same relationship as clearly (as the median of married status CATE is higher). This is not necessarily inconsistent, as the BLP reflects a conditional linear relationship after controlling for other covariate, whereas the boxplots only summarize the distribution of predicted CATEs. We therefore interpret marital status heterogeneity cautiously.

3.4 Targeting Operator Characteristic (TOC)

The TOC curve⁸ provides a summary of CF's ranking ability. It sorts each person's predicted CATE from largest to smallest. At a given treatment percentage q , TOC measures how much larger the ATE is among the top q of individuals ranked by predicted CATE, relative to the population average effect from treating a random fraction of the same size. Therefore, it answers a policy-related question, when budget is fixed, and it only allows treatment for only part of the population, how much we can benefit from prioritizing individuals with the highest predicted CATEs? If the TOC curve is declining and its confidence interval lies entirely above zero, the forest ranking has meaningful targeting value, indicating that individuals ranked at the top benefit more from treatment than a randomly selected group of the same size. If the curve is declining and CI is entirely above 0, forest ranking has targeting value that sample ranked the front will benefit more.

In previous sections, we have shown evidence of the heterogeneity, but BLP can not indicate whether the heterogeneity is large enough to support heterogeneous policy design. But TOC can fill this gap by transforming CATE distribution to a specific targeting profit function. Figure A.7 shows the comparable results of both enactment and implementation effect. During the enactment period, the curve is flat with the CI line covering zero, indicating that this heterogeneity can not transform into strong targeting or ranking power. During the implementation period, the curve demonstrates a declining trend, from around 0.22 at $q=0.1$ to 0 at $q=1$, with CI entirely above 0. The top 10% of individuals ranked by predicted CATE experience an average effect many times (10-20) the population ATE, confirming that the identified heterogeneity has sufficient magnitude to support meaningful policy targeting. In addition, the heterogeneity policy value appears gradually over time. During enactment, it only has direction but no magnitude. Only during implementation does it have both direction and sufficient magnitude to support targeting. Taken together, they are conclusions that the original paper's simple ATE framework cannot draw.

4 Age Placebo Test

The robustness checks in the original paper include a time placebo test that fixes the treatment age group at [19,25] and varies the policy date across 17 pre-enactment months. We complement this with an age placebo that fixes the policy date and varies the treatment age window across all possible 6-year intervals. In addition, we use a permutation placebo test because it does not rely on large-sample asymptotic approximations and instead evaluates whether the estimate for the true treatment age window is large relative to the distribution generated by alternative placebo age windows.

⁸I got this method to test policy target from the official website of *grf* package that constructs causal forest in R. See link.

Figure A.8 and A.9 demonstrate the estimated policy effects are specific to the legally mandated age window [19,25] across all five outcomes. The left column is the baseline placebo test as we switch the treatment age group, while the right column shows the permutation test results. For *anyhi*, the true policy window produces the greatest coverage improvement, but nearby windows may also show positive effects because *anyhi* is a general net insurance measurement, and those windows substantially overlap with the actually treated ages⁹. The permutation test also shows a randomization p-value of 0.000. For *emphi_dep*, the evidence is similar but weaker, and it is therefore better to interpret as suggestive (randomization $p=0.063$). For other outcomes *indiv* and *emphi*, the true window produces the most negative effects, consistent with young adults switching away from individually purchased and own-name employer coverage towards parental ESI. For *govhi*, the true effect is insignificant, confirming that the ACA mandate did not significantly affect government-provided coverage for any age group.

5 Policy Suggestion: Take-up Friction Analysis¹⁰

The heterogeneity analysis in the previous section has shown that the treatment effects vary across the sample. Therefore, a question comes up naturally: among all youngsters who can access parental ESI directly, as well as for those who remains uninsured despite their eligibility, which factors will prevent take-up? The original paper discussed it in Table 6, but it stopped at descriptive stage, since it compared observable characteristics of takers and non-takers using bivariate t-tests. However, this approach cannot capture the joint prediction power of multiple factors, and it can not rank the relative importance of the different friction types. Therefore, we try to enrich the results by modernizing it via a random forest classifier (because the outcomes are 0/1 variables), identifying the most predictive factor(s) of non-takeup, as the basic logic is similar to that in Section 3.

5.1 Sample Construction and Non-takers

Following the original paper, we restrict the sample to youngsters with parental ESI (*emphi_parent*=1) in the post-policy period, corresponding to ones for whom the mandate creates a direct and coverage option with lower cost. Within this subsample, we define non-takers as who remain uninsured from any source (*anyhi*=0), the group who fails to take up is due to the frictions rather than limitations¹¹.

5.2 Random Forest Classifier

Technically, we train a random forest classifier using socioeconomic covariates at SIPP, including education, race, marital status, student status, income, self-reported health status, prior insurance status, and co-residence with parents, to predict non-takeup probability. To address the unbalance problem in the sample (since non-takers only account for 17%), we apply inverse-frequency class weights, which prevent the classifier from defaulting to the majority class and ensure that the non-taker group is adequately represented in the splitting criterion.

The results is shown in Figure A.11, demonstrating a clear level of the take-up frictions, indicating which factors are more predictive the non-takeup frictions than others. Pre-policy insurance status (*anyhi_pre*) is the strongest predictive factor since it ranks the first among others. This finding suggests that insurance inertia as the strongest predictor(Handel (2013); Ericson (2014)): youngsters who have already held coverage from another source in the pre-policy period are most likely to remain outside of the ESI post-mandate, because the default of existing coverage creates a behavioral obstacle to switch to another source of insurance.

⁹Therefore, the age placebo test should be interpreted as a concentration test rather than a falsification test.

¹⁰This section uses a modernized model to extend analysis that did not discussed in detail. There will be no direct results from the original paper to support the findings in this section.

¹¹This definition is stricter than a simple filter: *emphi_dep*=0, since it eliminated individuals who chose any other sources of insurance

Hispanic ethnicity ranks in the second place, consistent with information-related and language barriers that limits policy awareness and enrollment capacity among this group. In addition, full-time student status also ranks among the top tier, suggesting that student status or related insurance remains an important factor of take-up behavior.

Following the top tier, education-related variables, including high-school drop out and income level (*fpl_ratio*) follows the 1st-tier predictor. Notably, physical co-residence with parents (*live_wparent*) ranks near the bottom of the importance, indicating that logistical proximity is less binding than information-related obstacles once other socioeconomic factors are controlled, which challenges the intuition that geographic access is a primary factor.

These findings are largely consistent with the original paper’s Table 6. The original paper documents that uninsured but eligible youngsters are more likely to be Hispanic, Black, less educated, lower-income, and in worse health condition relative to those who took up parental coverage. Our random forest variable importance rankings agree with the following patterns: Hispanic and African ethnicity, education variables (*hsdo*, *hsg*), and income (*fpl_ratio*) all rank among the top predictors of non-takeup.

5.3 Calibration

In this section, we use a calibration plot to assess whether the model can predict non-takeup probabilities are convincing. Each dot represents a decile of individuals ranked by predicted risk, plotted against the actual non-takeup rate. A perfect calibration will exactly align with the 45-degree line.

The calibration result is shown in Figure A.10. It presents clear rank order that non-takeup rates increase monotonically from 4% to 46% in the highest decile. The blue curve connecting them closely track the 45-degree line, especially in the middle decile ranges, indicating a well-calibrated predictions. In terms of the extreme value, the model slightly underestimates risk in the lowest decile, while there’s an overestimation in the highest decile. For policy suggestions, rank ordering rather than calibration is more important since the model can identify who is at highest risk, providing a basis for targeted applications.

5.4 Policy Suggestion

These findings carry direct implications for future policy design. The dominance of insurance inertia (*anyhi_pre*) as a predictor suggests that interventions targeting individuals who already hold coverage from other sources, and who may perceive the ACA mandate as redundant given their existing coverage, could substantially reduce the remaining coverage gap. We suggests that outreach efforts may be most productive when targeted toward individuals who already hold any other coverage. Hispanic ethnicity ranked the second, indicating the importance of cultural factors: the original paper shows that Hispanic youngsters are disproportionately represented among those who remain uninsured despite parental ESI access, and our variable importance analysis confirms that this is not merely a correlate of lower income or education, but an independent predictor of non-takeup frictions.

The relative unimportance of physical coresidence (*live_wparent*) suggests that digital outreach may be as effective as in-person assistance, since proximity to parents is not the binding constraint for most non-takers. Taken together, these results translate the original paper’s descriptive profile of the uninsured but eligible population into a more targeted prediction framework for external validity.

6 Conclusion and Discussions

This modernization report doesn’t try to overturn the original results. We also confirm that ACA mandate had significantly increase the parental employer sponsored insurance take-up among youngsters.

Furthermore, we enrich the paper’s finding in the following dimensions. Firstly, DRDID and DRDDD indicate that, the main take-up results (variable *anyhi* and *emphi_dep*) remain robust beyond linear models (LPM), indicating that the previous analyses did not affected by the form of the models. Secondly, event study and HonestDiD conduct a more rigorous assessment of the identification assumption, especially parallel trend. Although sensitivity check triggered some issue in distant pre-policy periods, the results remain convincable if we restrict the nearer period relative to ACA implementation. Third, the age placebo test strengthens the design by showing that the estimated effects are concentrated in the legally relevant treatment age range.

The main substantive enrichment stems from the heterogeneity and non-takeup analyses. The causal forest shows that the policy response can not be only summarized by a single ATE: most heterogeneity dimension becomes more salient during implementation, and the take-up response varies systematically with education, prior dependent coverage, marital status, and student status. In addition, we propose future policy targeting by random forest analyses, further extending the paper by showing that non-takeup among eligible young adults is predictable using observable characteristics, which points to the value of targeted outreach. At the same time, these enrichment should be interpreted with caution. The causal forest specifications rely on transformed cross-sectional constructions rather than the paper’s original DiD estimator, and the random forest results identify predictors of non-takeup rather than causal mechanisms, which may lose within-person variation in this process. Overall, the modernization extends the original paper’s main conclusion intact, but make it more credible, more nuanced, more policy-relevant, and more suggestive.

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Appendix

A Additional Figures and Tables

Table A.1: DRDID vs. Original LPM DiD Estimates

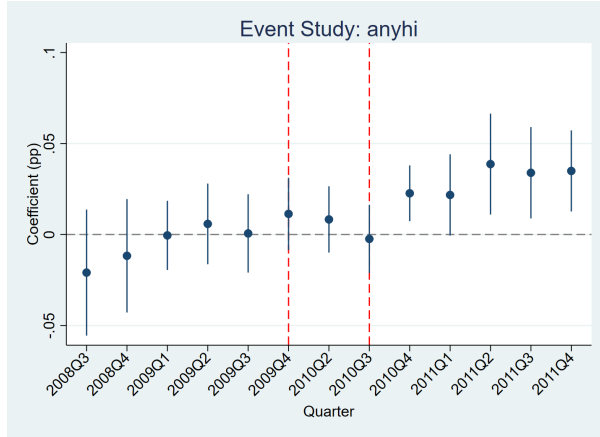
	Any Source		Emp. Dep.		Indiv.		Own ESI		Gov.	
	DR	LPM	DR	LPM	DR	LPM	DR	LPM	DR	LPM
Panel A: Enactment Effect (Mar-Sep 2010)										
DID	0.1176*** (0.0283)	-0.0018 (0.0064)	0.0631** (0.0322)	0.0239*** (0.0055)	-0.0058 (0.0191)	0.0025 (0.0031)	-0.0091 (0.0249)	-0.0173*** (0.0053)	0.0407** (0.0194)	-0.0106* (0.0054)
Panel B: Implementation Effect (Oct 2010-Nov 2011)										
DID	0.1143*** (0.0279)	0.0318*** (0.0074)	0.0733** (0.0316)	0.0702*** (0.0069)	-0.0186 (0.0164)	-0.0080*** (0.0023)	0.0030 (0.0276)	-0.0312*** (0.0060)	0.0439* (0.0250)	-0.0025 (0.0058)

Notes: DRDID estimates are Average Treatment Effects on the Treated (ATET) from the doubly robust DiD estimator of Sant'Anna and Zhao (2020), estimated on cross-sectional data collapsed to one observation per individual per period. LPM estimates are from the original linear probability model DiD regression in Table 2 of the original paper. Standard errors are reported in parentheses. DRDID standard errors are bootstrapped, while LPM standard errors are clustered at the state level. The *anyhi* estimates from DRDID are reported for completeness but should be interpreted cautiously. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$, hereafter.

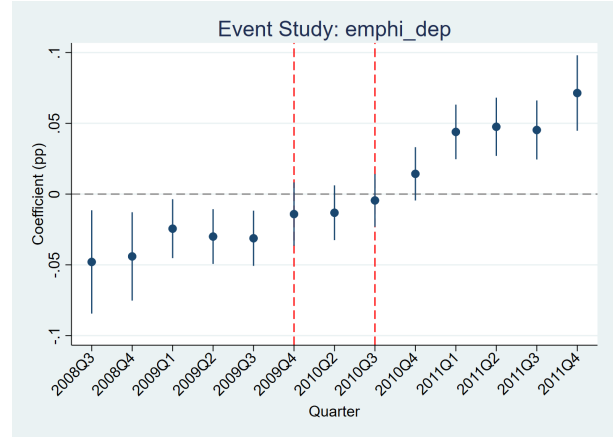
Table A.2: Causal Forest ATE Results

	Any Source		Emp. Dep. Coverage	
	DID ATE	CF ATE	DID ATE	CF ATE
CF1: Last available observation				
Enactment			0.0239*** (0.0055)	0.005 (0.007)
Implementation			0.0702*** (0.0069)	0.019*** (0.009)
CF2: Average difference between pre- and post-policy periods				
Enactment	-0.0018 (0.0064)	0.013 (0.016)		
Implementation	0.0318*** (0.0074)	0.016* (0.011)		

Notes: In this table, “DiD ATE” reports the baseline estimates from the original paper’s linear probability model specification. “CF ATE” reports causal forest ATE from the transformed cross-sectional settings used in this paper. CF1 uses the last available post-period observation for employer dependent coverage (*emphi_dep*), while CF2 uses the average pre-post difference outcome for any insurance (*anyhi*). Standard errors are reported in parentheses.

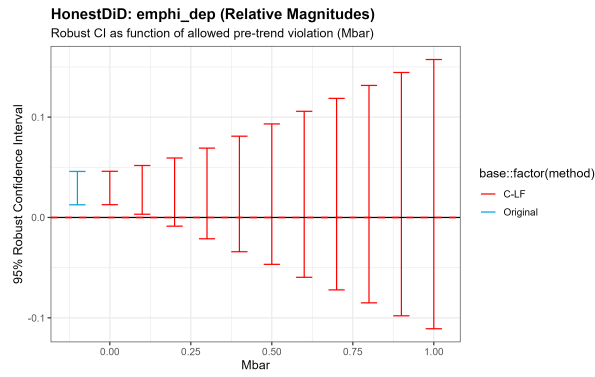


(a) Event Study: Any Source

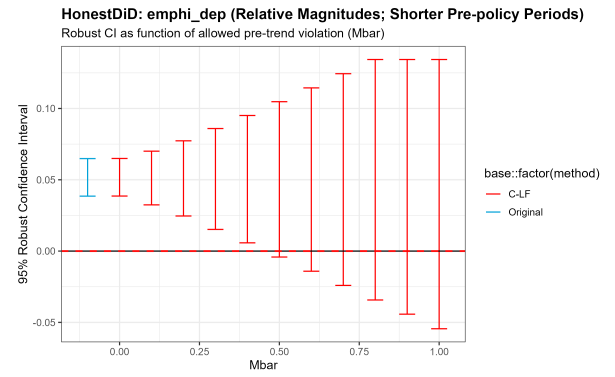


(b) Event Study: Employer Dependent Coverage

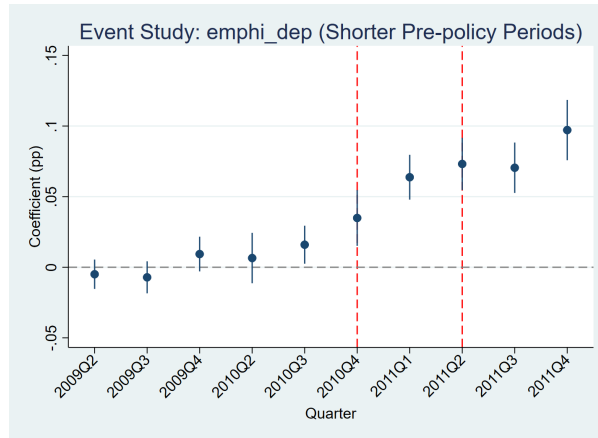
Figure A.1: Event Study



(a) Honest DiD: All Pre-period



(b) Honest DiD: Restricted Pre-period



(c) Event Study (*emphi_dep*): Restricted Pre-period

Figure A.2: Honest DiD and Restricted Event Study Result

Table A.3: DRDDD vs. Original DDD Results

	Any Source	Emp. Dep. Coverage	Indiv. Purchased	Own-Name ESI	Gov. Provided
DDD	0.0669*** (0.0235)	0.0961*** (0.0104)	-0.0131* (0.0067)	-0.0245 (0.0147)	0.0132 (0.0216)
DRDDD (Treatment)	0.082*** (0.020)	0.148*** (0.023)	-0.015 (0.010)	-0.045** (0.022)	-0.015 (0.015)
DRDDD (Falsification)	-0.053 (0.036)	N/A N/A	0.004 (0.011)	0.024 (0.026)	-0.059* (0.031)
DRDDD (Difference)	0.135*** (0.041)	0.148*** (0.023)	-0.019 (0.034)	-0.069** (0.034)	0.044 (0.031)

Notes: Original DDD estimates are taken from Table 5 of the original paper. However, the falsification estimate for *emphi_dep* is exactly zero with an undefined standard error, indicating a structural feature: dependent employer coverage is unavailable to young adults whose parents were without ESI, so the outcome variable exhibits zero variation in this subsample by construction. The last estimator is the difference of DRDDD treatment and falsification estimation, with standard errors calculated by the square root of the sum of squared subgroup standard errors (except for column *emphi_dep*).

Table A.4: Causal Forest Heterogeneity Results

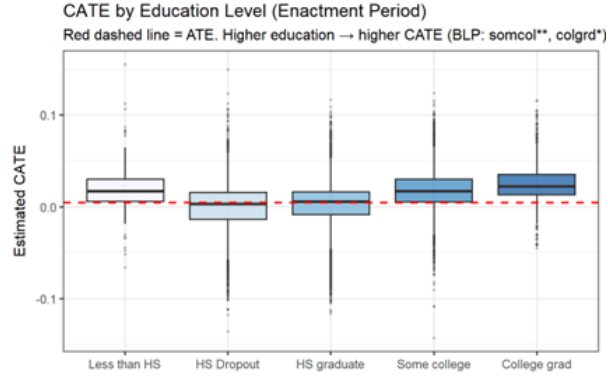
Variable	CF1 Enactment	CF1 Implementation
HS Dropout	***	***
HS Graduate	***	***
Some College	**	ns
College Grad	*	ns
Black	*	ns
Married	*	***
Female	ns	ns
Student	ns	**
White	ns	ns

Notes: Table reports Best Linear Projection (BLP) significance levels for selected covariates from the causal forest heterogeneity analysis. The BLP projects individual-level CATE estimates onto observable characteristics, testing which covariates linearly can mostly predict treatment effect heterogeneity. CF1 retains each individual's last observed post-period outcome (cross-sectional specification). The outcome variable is *emphi_dep* for all specifications. Significance levels are based on one-sided heteroskedasticity-robust (HC3) standard errors. ns = not statistically significant. For simplicity, we only listed representative covariates that are always significant among enactment and implementation period.

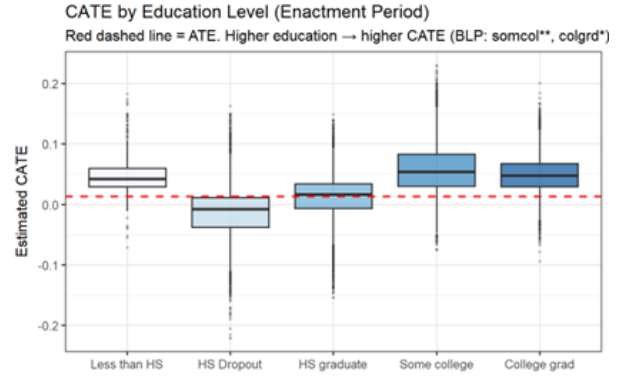
Table A.5: Best Linear Projection of Causal Forest Predictions (Calibration Test)

Prediction	CF1 Enactment	CF1 Implementation	CF2 Implementation
ATE	0.723 (0.609)	1.061*** (0.124)	0.838*** (0.195)
Heterogeneity	0.541** (0.225)	1.126*** (0.076)	1.182*** (0.139)

Notes: This table reports calibration test results following Chernozhukov et al. (2018), implemented via *test_calibration()* in the *grf* package. ATE row tests whether the forest's mean prediction is correctly scaled (mean.forest.prediction); a coefficient near 1.0 indicates accurate average effect estimation. Heterogeneity row tests whether individual CATE rankings contain genuine variation beyond the ATE (differential.forest.prediction); a significant coefficient rejects the null of treatment effect homogeneity. The outcome variable is *emphi_dep* for all specifications. Standard errors in parentheses are one-sided heteroskedasticity-robust (HC3).

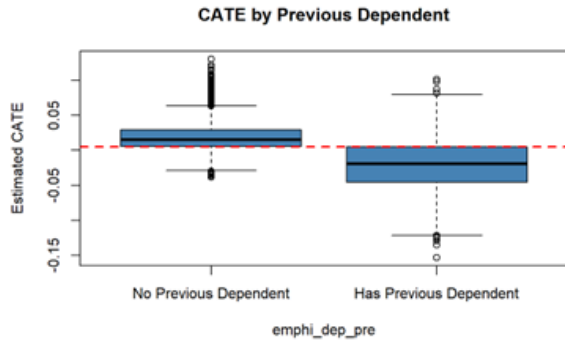


(a) Enactment Period

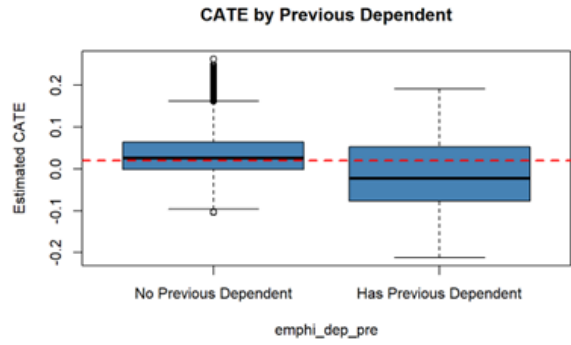


(b) Implementation Period

Figure A.3: Education Heterogeneity

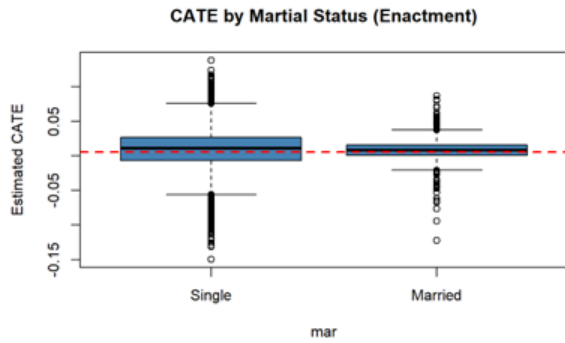


(a) Enactment Period

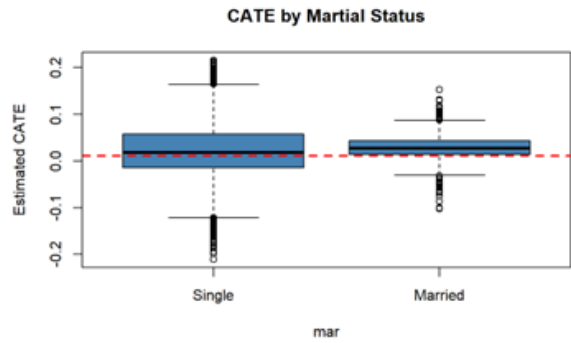


(b) Implementation Period

Figure A.4: Previous Dependent Heterogeneity

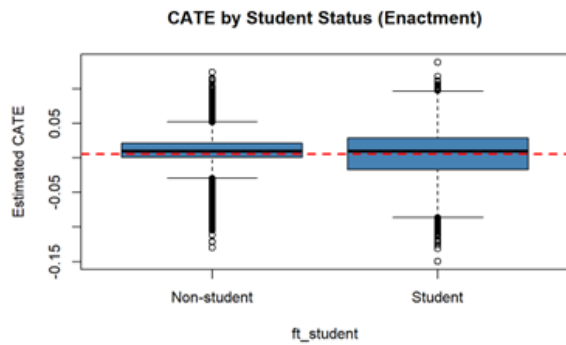


(a) Enactment Period

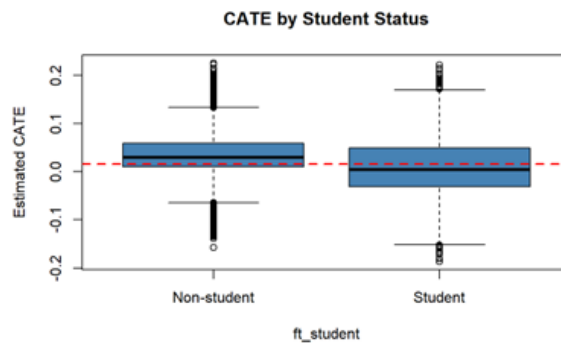


(b) Implementation Period

Figure A.5: Marital Status Heterogeneity

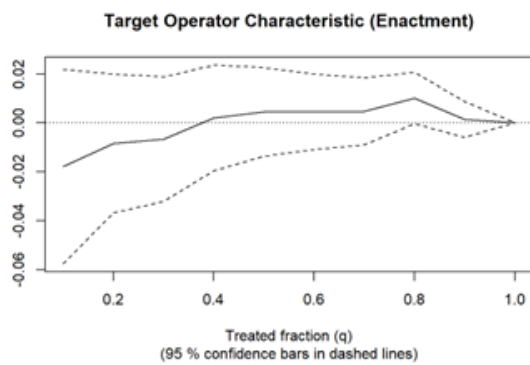


(a) Enactment Period

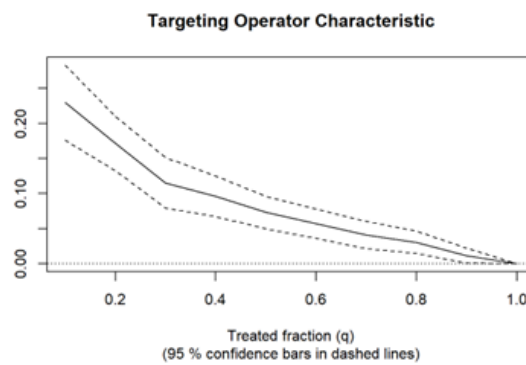


(b) Implementation Period

Figure A.6: Student Status Heterogeneity



(a) Enactment Period



(b) Implementation Period

Figure A.7: TOC Curve

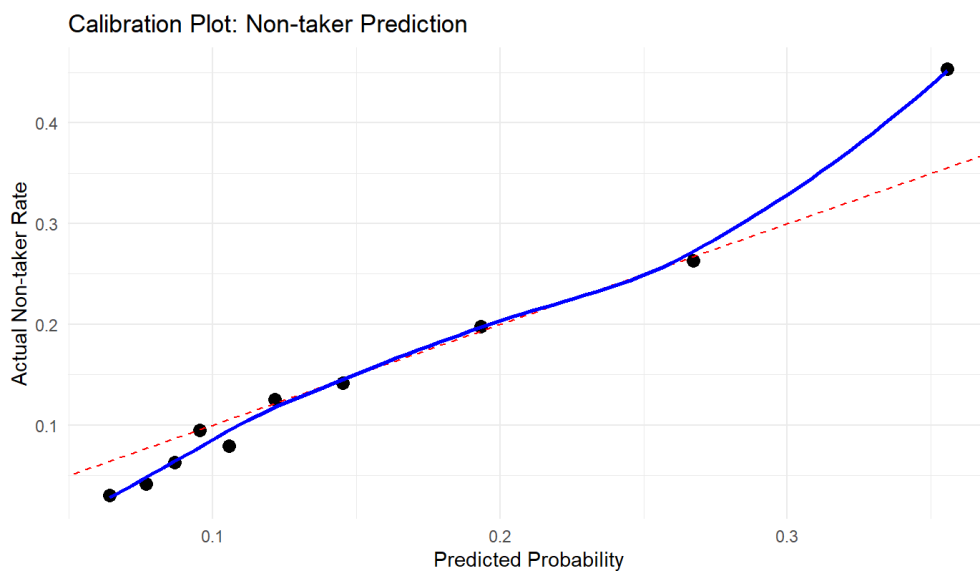


Figure A.10: Calibration

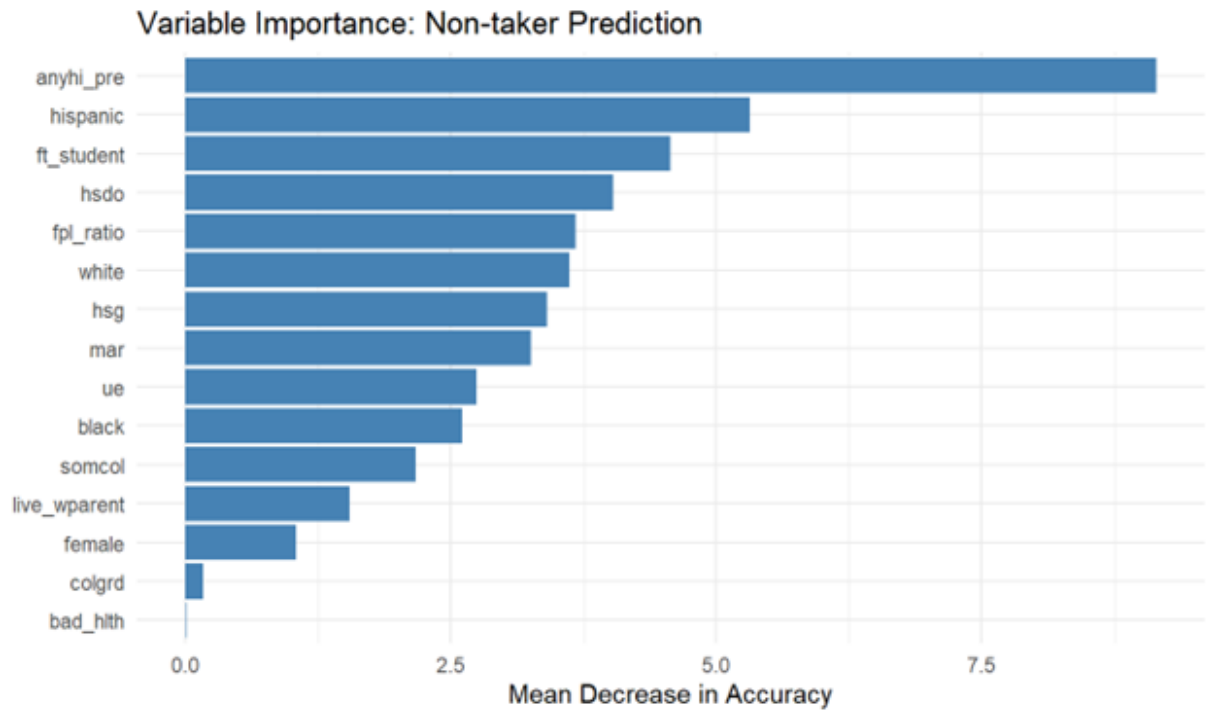
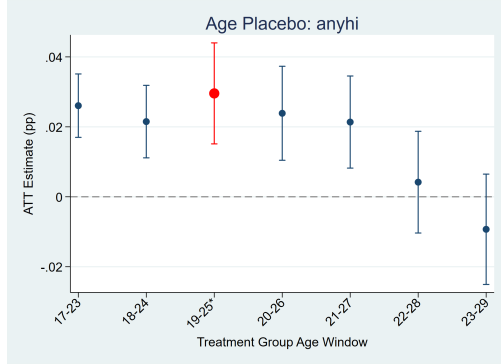
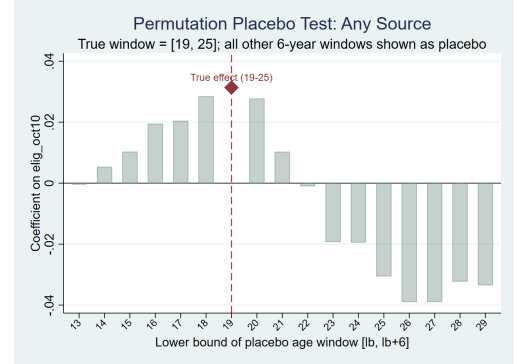


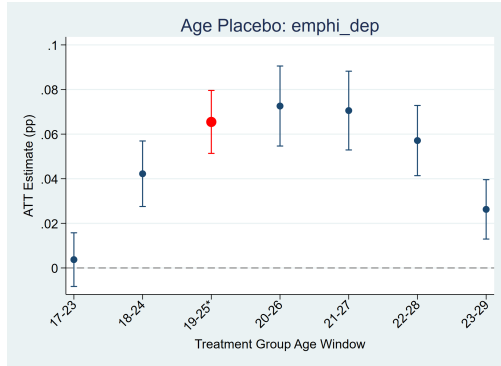
Figure A.11: Variable Importance for Take-ups



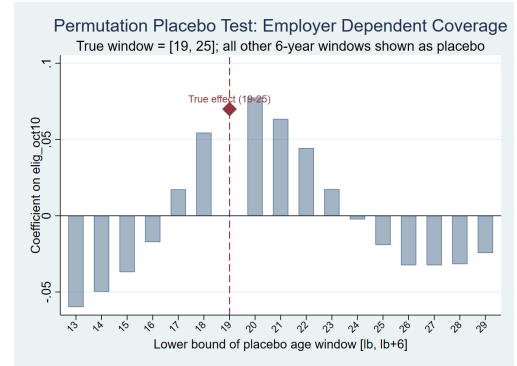
(a) Placebo Test: Any Source



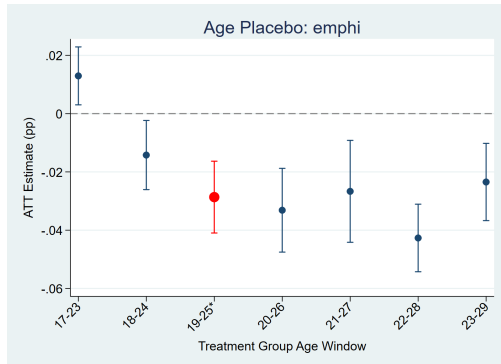
(b) Permutational Placebo Test: Any Source



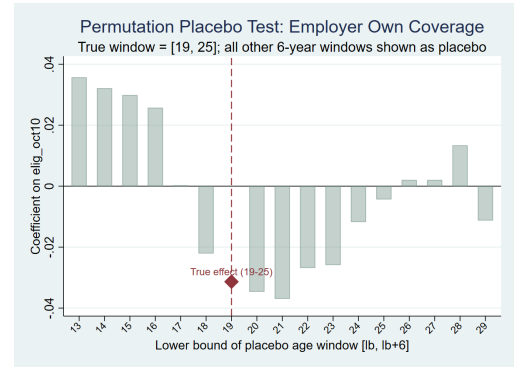
(c) Placebo Test: Employer Dependent Coverage



(d) Permutational Placebo Test: Employer Dependent Coverage

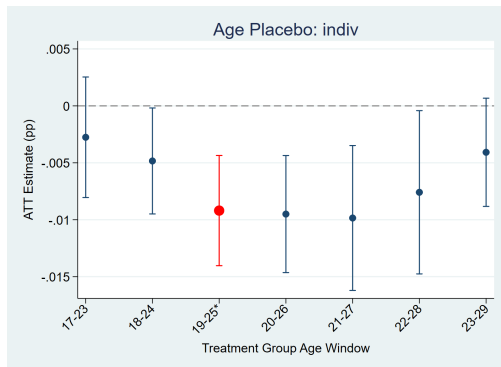


(e) Placebo Test: Employer Own Coverage

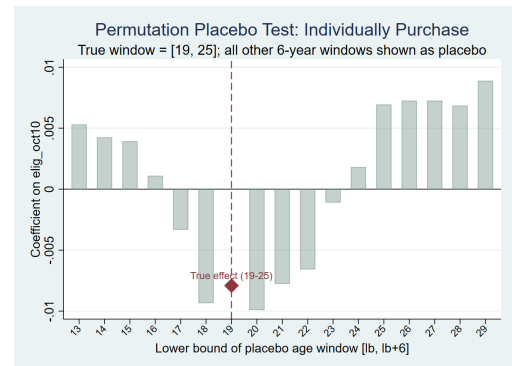


(f) Permutational Placebo Test: Employer Own Coverage

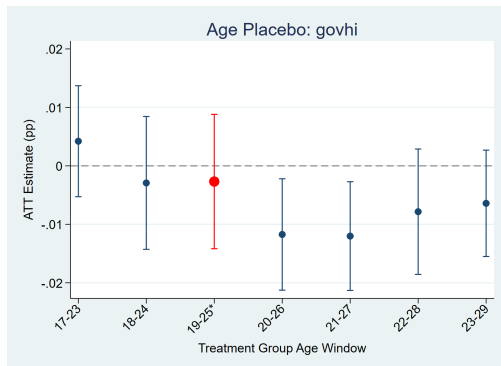
Figure A.8: Baseline and Permutational Placebo Test Result



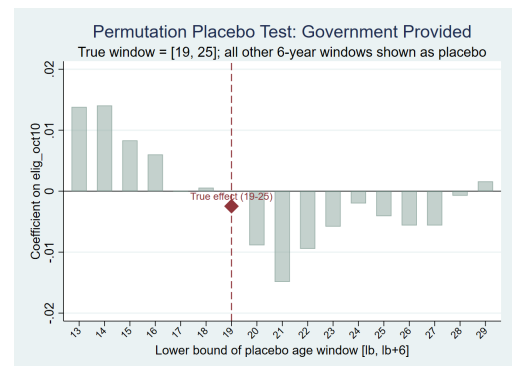
(a) Placebo Test: Individual Purchase



(b) Permutational Placebo Test: Individual Purchase



(c) Placebo Test: Government Provided



(d) Permutational Placebo Test: Government Provided

Figure A.9: Baseline and Permutational Placebo Test Result (Continued)